



1. Description

1.1. Project

Project Name	Seeed-LoRa-E5
Board Name	custom
Generated with:	STM32CubeMX 6.16.1
Date	01/18/2026

1.2. MCU

MCU Series	STM32WL
MCU Line	STM32WLEx
MCU name	STM32WLE5JCIx
MCU Package	UFBGA73
MCU Pin number	73

1.3. Core(s) information

Core(s)	ARM Cortex-M4
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2. Pinout Configuration



UFBGA73 (Top view)

3. Pins Configuration

Pin Number UFBGA73	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
A1	VSSSMPS	Power		
A2	VDDSMPS	Power		
A4	PA14	I/O	DEBUG_JTCK-SWCLK	
A5	VDDA	Power		
A7	VDD	Power		
A8	VBAT	Power		
B1	VLXSMPS	Power		
B2	VFBSMPS	Power		
B6	PC14-OSC32_IN	I/O	RCC_OSC32_IN	
B7	VSS	Power		
B8	PA13	I/O	DEBUG_JTMS-SWDIO	
C1	PB3 *	I/O	GPIO_Output	DBG3
C2	PB4 *	I/O	GPIO_Output	DBG4
C3	PB7	I/O	USART1_RX	USARTx_RX
C5	PC15-OSC32_OUT	I/O	RCC_OSC32_OUT	
D2	PB5 *	I/O	GPIO_Output	LED2
D6	PA0 *	I/O	GPIO_Output	DBG1
D7	PB13	I/O	GPIO_EXTI13	BUT1
D9	VSS	Power		
E1	PB6	I/O	USART1_TX	USARTx_TX
E2	VDD	Power		
E3	VSS	Power		
E8	VDDRF	Power		
E9	VDD	Power		
F5	NRST	Reset		
F6	PB0-VDD_TCXO	I/O	VDDTCXO	
F7	VDDRF1V55	Power		
G5	VSS	Power		
G6	VSSRF	Power		
G7	VSSRF	Power		
G8	VSSRF	Power		
G9	OSC_IN	MonoIO	RCC_OSC_IN	
H4	PB10 *	I/O	GPIO_Output	DBG2
H5	VDD	Power		
H6	VSSRF	Power		
H7	RFI_N	MonoIO		

Pin Number UFBGA73	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
H8	VDDPA	Power		
H9	VR_PA	Power		
J1	PA4 *	I/O	GPIO_Output	RF_CTRL1
J2	PA5 *	I/O	GPIO_Output	RF_CTRL2
J6	RFI_P	MonolIO		
J8	RFO_LP	MonolIO		
J9	RFO_HP	MonolIO		

* The pin is affected with an I/O function



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32WL
Line	STM32WLEx
MCU	STM32WLE5JCIx
Datasheet	DS13105 Rev7

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(AAA700)
Capacity	700.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	10.0 mA
Max Pulse Current	30.0 mA
Cells in series	1
Cells in parallel	1

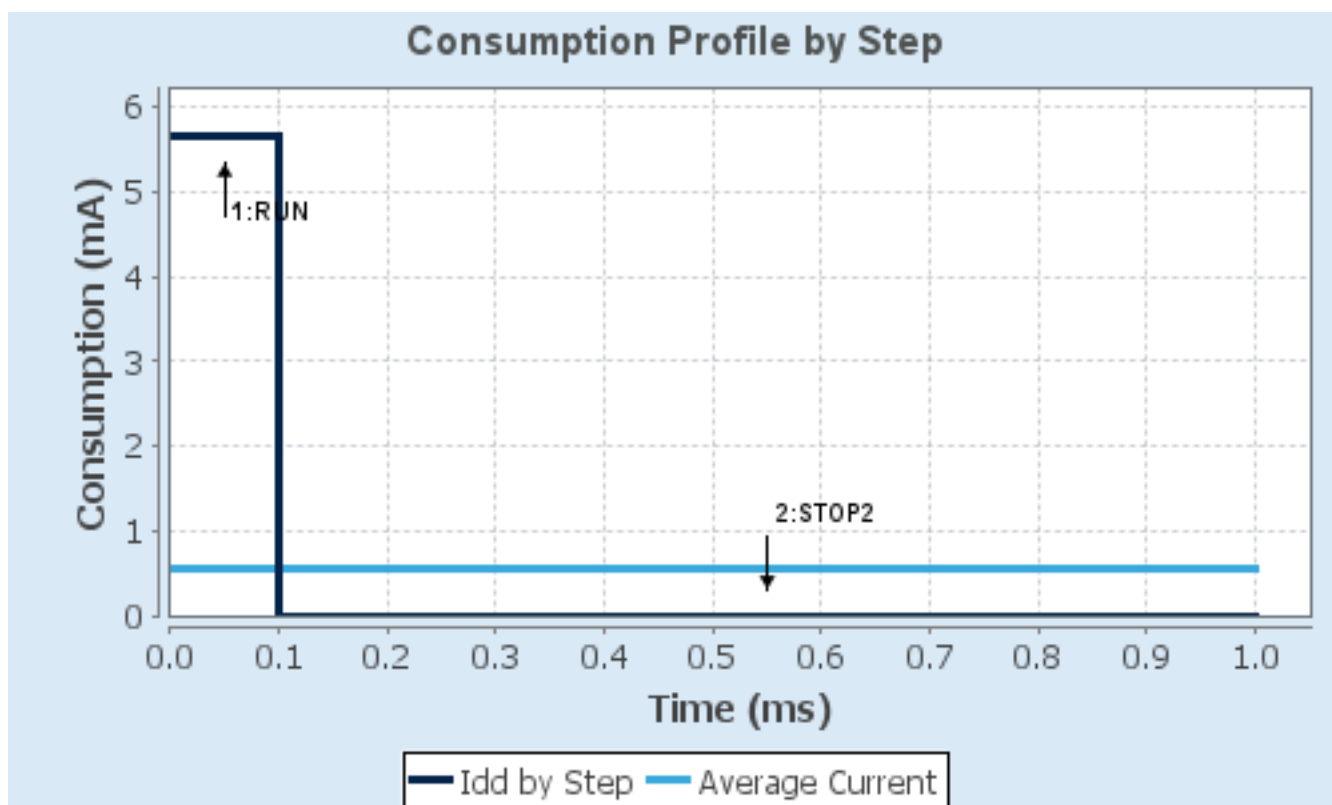
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP2
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-Medium/SMPS-OFF	NoRange
Fetch Type	SRAM1	NA
CPU Frequency	48 MHz	0 Hz
Clock Configuration	MSI	ALL_CLOCKS_OFF
Clock Source Frequency	48 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	5.65 mA	885 nA
Duration	0.1 ms	0.9 ms
DMIPS	60.0	0.0
Ta Max	124.47	125
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	565.8 μ A
Battery Life	1 month, 21 days, 1 hour	Average DMIPS	60.0 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	Seeed-LoRa-E5
Project Folder	C:\Users\danam\STM32CubeIDE\Alt-Workspace\Seeed-LoRa-E5
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_WL V1.4.0
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_RTC_Init	RTC
5	MX_ADC_Init	ADC
6	MX_USART1_UART_Init	USART1
7	MX_SUBGHZ_Init	SUBGHZ
8	MX_LoRaWAN_Init	LORAWAN

3. *Peripherals and Middlewares Configuration*

3.1. ADC

mode: Temperature Sensor Channel

mode: Vrefint Channel

3.1.1. Parameter Settings:

ADC_Settings:

Clock Prescaler

Resolution

Calibration

Data Alignment

Scan Conversion Mode

End Of Conversion Selection

Low Power Auto Wait

Auto Off

Continuous Conversion Mode

Discontinuous Conversion Mode

External Trigger Conversion Source

External Trigger Conversion Edge

DMA Continuous Requests

Overrun behaviour

Sequencer

SamplingTime Common 1

SamplingTime Common 2

Oversampling Mode

Trigger Frequency

ADC_Regular_ConversionMode:

Enable Regular Conversions

Synchronous clock mode divided by 4 *

ADC 12-bit resolution

Disable

Right alignment

Disabled

End of single conversion

Disabled

Disabled

Disabled

Disabled

Regular Conversion launched by software

None

Disabled

Overrun data overwritten *

Sequencer set to fully configurable

160.5 Cycles *

160.5 Cycles *

Disabled

High frequency

Disable

3.2. ADV_TRACE

mode: Enabled

3.3. DEBUG

JTAG and Trace: Serial Wire

3.4. MISC

mode: Enabled

3.5. RCC

High Speed Clock (HSE): TCXO

Low Speed Clock (LSE) : Crystal/Ceramic Resonator

3.5.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Disabled
Data Cache	Enabled
Flash Latency(WS)	2 WS (3 CPU cycle)

RCC Parameters:

HSI Calibration Value	64
MSI Calibration Value	0
MSI Auto Calibration	Enabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
LSE Drive Capability	LSE oscillator low drive capability

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
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3.6. RTC

mode: Activate Clock Source

mode: Activate Calendar

Alarm A: Internal Alarm A

3.6.1. Parameter Settings:

General:

Asynchronous Predivider value
Bin Mode
SSRU Underflow Interrupt

RTC_PREDIV_A *

Free running Binary mode *

Enabled

Alarm A:

Free running 32 bit value
Binary AutoControl
Free running 32 bit mask

0

RTC_ALARMSUBSECONDBIN_AUTOCLR_NO *

SS[31:0] are compared and must match to activate alarm

3.7. SEQUENCER

mode: Enabled

3.8. SUBGHZ

mode: Activated

3.8.1. Parameter Settings:

Baudrate Prescaler Value **4 ***

3.9. SYS

Timebase Source: None

3.10. TIMER

mode: Enabled

3.11. TINY_LPM

mode: Enabled

3.12. USART1

Mode: Asynchronous

3.12.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Enable *
Txfifo Threshold	1 eighth full configuration

Rxfifo Threshold 1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.13. LORAWAN

mode: Enabled

3.13.1. LoRaWAN application:

Application selection:

Application

Application configuration recommendations

End Node skeleton *

!! Please read carefully Information panel below!!

board settings:

Send Tx on Timer or Button Evt

Probes Lines in Platform Settings

TX_ON_TIMER

false

lora_app:

Active region

Transmission duty cycle

Application user port

Switch class port

Default class

Default handler message state

Handler Adaptive Data Rate

Default activation type

Force rejoin at each reboot

Default Tx output power

Default Unicast ping slots periodicity

Default reponse timeout for class b and class c confirmed downlink frames in milli seconds.

LORAMAC_REGION_US915 *

10000

2

3

CLASS_A

Unconfirmed message

On

OTAA

true

TX_POWER_0

4

8000

sys_conf:

Trace verbose level

Enable Application Logging

Disable Low Power Mode

Enable Sensor

VLEVEL_M

true

false

false

3.13.2. LoRaWAN commissioning:

Commissioning:

Public network	true
Current network ID	0
se-identity:	
LoRaWAN device EUI	00,00,00,00,00,00,00,00
App/Join EUI	01,01,01,01,01,01,01,01
Application key	2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C
Network key	2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C
Device Address	00,00,00,00
Network session key	2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C
Application session key	2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C
lorawan_conf:	
Enable Key read access	true

3.13.3. LoRaWAN middleware:

lorawan_conf:

Region(s) selection	please select the desired region(s) in the list below
Region Asia freq: 923	false
Region Australia freq: 915	false
Region China freq: 470	false
Region China freq: 779	false
Region Europe freq: 433	false
Region Europe freq: 868	true
Region Korea freq: 920	false
Region India freq: 865	false
Region USA freq: 915	true
Region Russia freq: 864	false
Enable Hybrid mode	false
Enable LoRaMAC ClassB	false
Enable the context management storage	true *
Select the LoRaWAN Link Layer specification version	v1.0.4
Enable the additional LoRaWAN packages	true *
Select the LoRaWAN packages version	

radio_conf:		v2.0.0 *
Radio maximum wakeup time (in ms)		1
radio_board_if:		
Select radio Driver		User Board
TCXO support		true
DCDC support		true
Tx Rfo Config		CONF_RFO_HP *
mw_log_conf:		
Enable Middleware log		true

3.13.4. Platform Settings:

ADC	ADC
RTC	RTC
USART	USART1

* User modified value

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
DEBUG	PA14	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	
	PA13	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	
RCC	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a	
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	
	PB0-VDD_TCXO	VDDTCXO	n/a	n/a	n/a	
	OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
USART1	PB7	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	USARTx_RX
	PB6	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	USARTx_TX
GPIO	PB3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	DBG3
	PB4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	DBG4
	PB5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	LED2
	PA0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	DBG1
	PB13	GPIO_EXTI13	External Interrupt Mode with Falling edge trigger detection	Pull-up *	n/a	BUT1
	PB10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	DBG2
	PA4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	RF_CTRL1
	PA5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	RF_CTRL2

4.2. DMA configuration

DMA request	Stream	Direction	Priority
USART1_TX	DMA1_Channel1	Memory To Peripheral	Low

USART1_TX: DMA1_Channel1 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
RTC Tamper, RTC TimeStamp, LSECSS and RTC SSRU Interrupts	true	0	0
DMA1 Channel 1 Interrupt	true	2	0
USART1 Interrupt	true	2	0
EXTI Lines [15:10] Interrupt	true	0	0
RTC Alarms (A and B) Interrupt	true	0	0
SUBGHZ Radio Interrupt	true	0	0
PVD and PVM detector	unused		
FLASH (CFI) global Interrupt	unused		
RCC Interrupt	unused		
ADC Interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	false
RTC Tamper, RTC TimeStamp, LSECSS and RTC SSRU Interrupts	false	true	true
DMA1 Channel 1 Interrupt	false	true	true
USART1 Interrupt	false	true	true

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
EXTI Lines [15:10] Interrupt	false	true	true
RTC Alarms (A and B) Interrupt	false	true	true
SUBGHZ Radio Interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

Middleware									
LORAWAN									
System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Utilities
DMA	ADC	RTC	SUBGHZ				DEBUG		ADV_TRACE
GPIO			USART1						MISC
IVIC									SEQUENCER
RCC									TIMER
SYS									TIHY_LPM

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32wl_bsdl.zip
HW Models	https://www.st.com/resource/en/hw_model/stm32wl_reference_board_mb1720-xo_ios_shield.7z
IBIS models	https://www.st.com/resource/en/ibis_model/stm32wl_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32wl_svd.zip
System View Description	https://www.st.com/resource/en/svd/stm32wl-svd.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1720_bom.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1789_hp_bom.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1789_lp_bom.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1791_bom.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1792_bom.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1842-bom.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1720_bdp.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1720_manufacturing.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1789_hp_bdp.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1789_hp_manufacturing.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1789_lp_bdp.zip

Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1789_lp_manufacturing.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1791_bdp.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1791_manufacturing.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1792_bdp.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1792_manufacturing.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1842-bdp.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1842-manufacturing.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1720_schematics.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1789_hp_schematic.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1789_lp_schematic.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1791_schematic.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1792_schematic.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1842-schematic.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf

Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32cube-monitor-wireless-longrange_rfttest.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32wl5x-ex-product-line-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/beyond-the-wires-exploring-bluetooth-and-lorawan-connectivity.pdf
Flyers	https://www.st.com/resource/en/flyer/flnucleolrwan.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32wl5ex.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-generalpurpose-

timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5406-how-to-build-a-lora-application-with-stm32cubewl-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5409-stm32cube-mcu-package-examples-for-stm32wl-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5457-rf-matching-network-design-guide-for-stm32wl-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5480-how-to-build-a-sigfox-application-with-stm32cubewl-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5042-how-to-calibrate-the-hse-clock-for-rf-applications-on-stm32-wireless-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4894-how-to-use-eeeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5407-how-to-optimize-the-rf-board-layout-for-stm32wl5xex-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5886-guidelines-for-design-and-board-assembly-of-land-grid-array-packages-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5646-how-to-select-a-

32-mhz-hse-oscillator-for-stm32wl5xex-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcusmpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-with-stm32-mcus-stmicroelectronics.pdf

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